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Course: AP

Roll:CSB24069
CouseCode:CSBT215

```
import java.util.*;
import java.util.stream.*;

class Student {

    int id;
    String name;
    List<String> courses;
    Map<String,Integer> scores;

    Student(int id,String name){
        this.id=id;
        this.name=name;
        courses=new ArrayList<>();
        scores=new HashMap<>();
    }

    double getAverageScore() {
        return scores.values()
            .stream()
            .mapToInt(Integer::intValue)
            .average()
            .orElse(0);
    }
}

public class Student_performance {

    public static List<Student> getTopNStudents(List<Student>
students,int n){

        return students.stream()

.sorted(Comparator.comparingDouble(Student::getAverageScore).reversed()
)

        .limit(n)
        .collect(Collectors.toList());
    }
}
```

```

    public static Map<String,Double>
getAverageScorePerCourse(List<Student> students) {

    Map<String,List<Integer>> courseScores = new HashMap<>();

    for(Student s:students){
        for(String course:s.courses){

            int marks = s.scores.getOrDefault(course,0);

            courseScores
                .computeIfAbsent(course,k->new ArrayList<>())
                .add(marks);
        }
    }

    Map<String,Double> avgMap = new HashMap<>();

    for(String course:courseScores.keySet()){

        List<Integer> marks = courseScores.get(course);

        double avg = marks.stream()
            .mapToInt(Integer::intValue)
            .average()
            .orElse(0);

        avgMap.put(course,avg);
    }

    return avgMap;
}

    public static Set<String> getAllUniqueCourses(List<Student>
students) {

    return students.stream()
        .flatMap(s -> s.courses.stream())
        .collect(Collectors.toSet());
}

```

```
}

public static void main(String[] args) {

    List<Student> students = new ArrayList<>();

    // Student 1
    Student s1 = new Student(24012, "Anil");
    s1.courses.addAll(Arrays.asList("DS", "AP", "CAO", "TOC"));
    s1.scores.put("DS", 42);
    s1.scores.put("AP", 38);
    s1.scores.put("CAO", 35);
    s1.scores.put("TOC", 40);
    students.add(s1);

    // Student 2
    Student s2 = new Student(24013, "Sahil");
    s2.courses.addAll(Arrays.asList("DS", "AP", "CAO", "TOC"));
    s2.scores.put("DS", 30);
    s2.scores.put("AP", 44);
    s2.scores.put("CAO", 36);
    s2.scores.put("TOC", 41);
    students.add(s2);

    // Student 3
    Student s3 = new Student(24014, "Aryan");
    s3.courses.addAll(Arrays.asList("DS", "AP", "CAO", "TOC"));
    s3.scores.put("DS", 33);
    s3.scores.put("AP", 28);
    s3.scores.put("CAO", 39);
    s3.scores.put("TOC", 45);
    students.add(s3);

    // Student 4
    Student s4 = new Student(24015, "Aditya");
    s4.courses.addAll(Arrays.asList("DS", "AP", "CAO", "TOC"));
    s4.scores.put("DS", 37);
    s4.scores.put("AP", 40);
```

```

s4.scores.put("CAO",29);
s4.scores.put("TOC",48);
students.add(s4);

// Student 5
Student s5 = new Student(24016,"Kaushal");
s5.courses.addAll(Arrays.asList("DS","AP","CAO","TOC"));
s5.scores.put("DS",41);
s5.scores.put("AP",43);
s5.scores.put("CAO",34);
s5.scores.put("TOC",39);
students.add(s5);

// Top 3 students
List<Student> topStudents = getTopNStudents(students,3);

System.out.println("Top Students:");
for(Student s:topStudents){
    System.out.println(s.name+" Avg:"+s.getAverageScore());
}

// Course averages
System.out.println("\nAverage per Course:");
System.out.println(getAverageScorePerCourse(students));

// Unique courses
System.out.println("\nAll Courses:");
System.out.println(getAllUniqueCourses(students));
}
}

```

Output:

```

PS C:\Users\kaush\Desktop\imp\vscode\AP assignment\assignment_nr6,7,> cd "c:\Users\kaush\Desktop\imp\vscode\AP assignment\as
; if ($?) { java Student_performance }
Top Students:
Kaushal Avg:39.25
Anil Avg:38.75
Aditya Avg:38.5

Average per Course:
{CAO=34.6, TOC=42.6, AP=38.6, DS=36.6}

All Courses:
[CAO, TOC, DS, AP]
PS C:\Users\kaush\Desktop\imp\vscode\AP assignment\assignment_nr6,7,\assignment_nr6>

```

Analsize of Time complexity:

Analysis of Time Complexity

1) Course Average

- Traversing all student and courses $\rightarrow O(n \times m)$
- Calculating average for each course $\rightarrow O(n)$

$$T(n) = O(n \times m)$$

2) Sorting top n students

- Computing average score $\rightarrow O(n)$
- Sort student by average score $\rightarrow O(n \log n)$

$$T(n) = O(n \log n)$$

So, Total time Complexity

$$\begin{aligned} \therefore T(n) &= T(n)_1 + T(n)_2 \\ &= O(n \times m) + O(n \log n) \end{aligned}$$
